SECTION 4: SYMMETRIC VS ASYMMETRIC ENCRYPTION [10 MARKS]

8. Symmetric Encryption Vulnerability: (3 Marks)

Explain why symmetric encryption suffers from the "Key Distribution Problem" and describe how hackers exploit it:

9. Asymmetric Public/Private Key Encryption Scenario: (4 Marks)

Zayan wishes to send a confidential, encrypted contract to Nadia over the Internet using asymmetric encryption.

a) Which cryptographic key does Zayan use to encrypt the contract?

b) Which key does Nadia use to decrypt and read the contract? _____

c) Can an eavesdropper who intercepts the message decrypt it using Nadia's Public Key? Explain why: (2 Marks)

10. Digital Signatures & SSL/TLS: (3 Marks)

a) What two security properties does a Digital Signature provide? (2 Marks)

b) State the role of a Certificate Authority (CA) in issuing SSL certificates for websites: (1 Mark)